Question 1

M2L12

Let X = the number of tosses of 2 dice until you observe a sum of 11

X is a geometric random variable with a probability of success of $\frac{1}{18}$

The table below shows the outcomes of the two tosses. A sum of 11 occurs in two cases: rolling a 6 and a 5 or a 5 and a 6. There are 36 possible outcomes and 2 of them are 11, so the probability is $\frac{1}{18}$

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

The expected value of a geometric random variable is $\frac{1}{p}$. For this scenario that is $\frac{1}{1/18} = 18$.

Thus, we would expect to have to roll the dice 18 times before seeing a sum of 11. This should make intuitive sense as the probability is $\frac{1}{18}$, so in 18 rolls we would expect to see the outcome.

Question 2

M2L12

Review Sections 4.2.2.1 in Goldsman, D. & Goldsman, P. (2020). A First Course in Probability and Statistics. Lulu. [Download link]

It can be shown that the interarrival times of a Poisson process are all independent and identically distributed exponential random variables with parameter λ.

Theorem: Let X be the amount of time until the *first arrival* of a Poisson process with rate λ. Then $X \sim \text{Exp}(\lambda)$.

Proof: The cdf of $F(x)$ is

$$\begin{aligned} F(x) &= \text{P}(X \leq x) \\ &= 1 - \text{P}(\text{no arrivals in } [0, x]) \\ &= 1 - \frac{e^{-\lambda x}(\lambda x)^0}{0!} \quad (\text{the number of arrivals in } [0, x] \text{ is Pois}(\lambda x)) \\ &= 1 - e^{-\lambda x}. \quad \square \end{aligned}$$

Theorem: Amazingly, it can be shown (after a lot of work) that the interarrival times of a Poisson process are *all* iid Exp(λ)! See for yourself when you take an advanced stochastic processes course.

Since we have a Poisson process, the distribution of the interarrival time between the 5th and 6th arrivals is $\text{Exp}(\lambda = 2)$

Question 3

M2L11

Since X and Y are independent random variables, their covariance will equal 0. This will mean their correlation will also be 0, so they are uncorrelated.

Question 4

M2L12

The definition of a Poisson process satisfies the following:

Arrivals occur one at a time at rate λ

Independent increments

Stationary increments

Question 5

M2L12

Can also use LOTUS directly instead of finding $E[1/X]$ first.

$$E\left[\frac{2}{X} - 3\right] = \int_0^1 \left(\frac{2}{x} - 3\right)(4x^3)dx = \int_0^1 8x^2 - 12x^3$$

$$= \frac{8x^3}{3} - \frac{12x^4}{4}\bigg|_0^1 = \frac{8}{3} - \frac{12}{4} = -\frac{1}{3}$$

Question 6

M2L8

The limits of $1 \leq y \leq e$ on the PDF of y are due to the fact that $0 \leq x \leq 1$ and $y = e^x$

When $x = 0 \rightarrow y = e^0 = 1$

When $x = 1 \rightarrow y = e^1$

An alternate way of finding the PDF of Y is using the “Bonus Theorem” from the lesson.

We have $f_X(x) = 3x^2$, $Y = h(X) = e^X$, $h^{-1}(y) = \ell n(y)$

$$f_Y(y) = f_X(h^{-1}(y))\left|\frac{d}{dy}h^{-1}(y)\right|$$

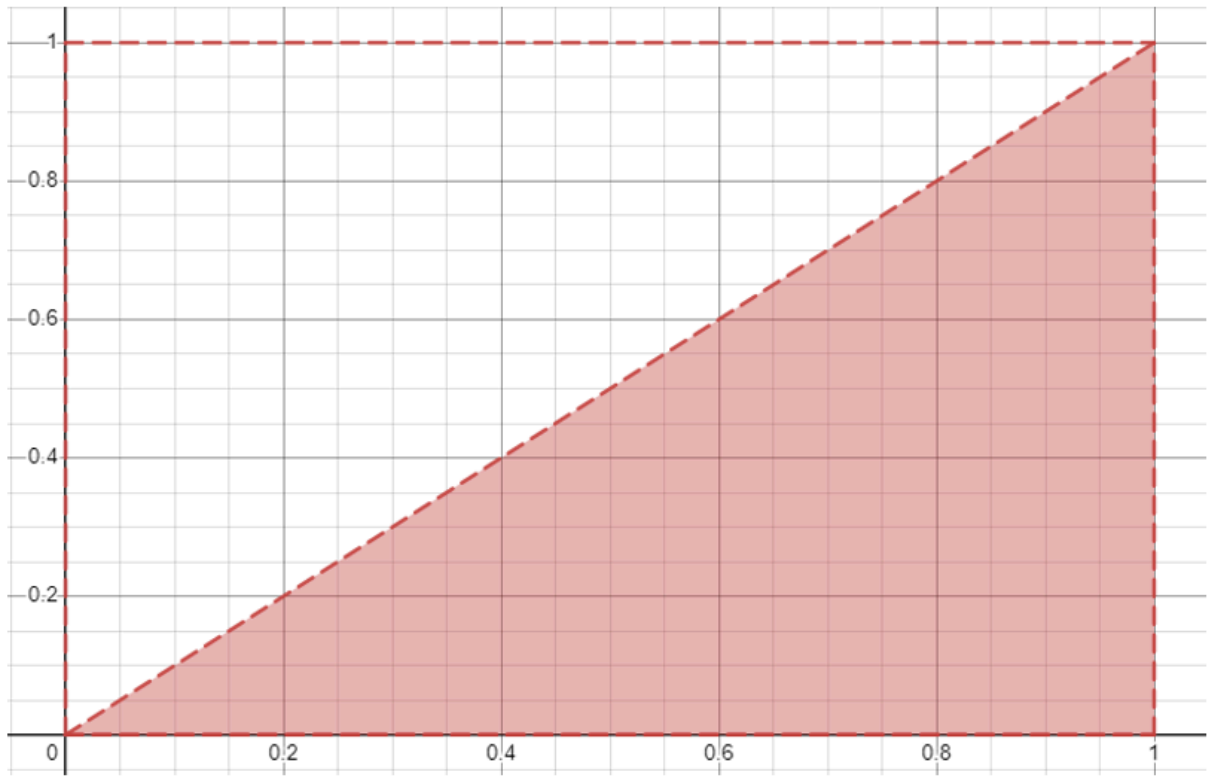
$$= f_X(\ell n(y))\left|\frac{d}{dy}\ell n(y)\right| = 3[\ell n(y)]^2\left|\frac{1}{y}\right| = \frac{3[\ell n(y)]^2}{y}$$

This matches the result from the solution document.

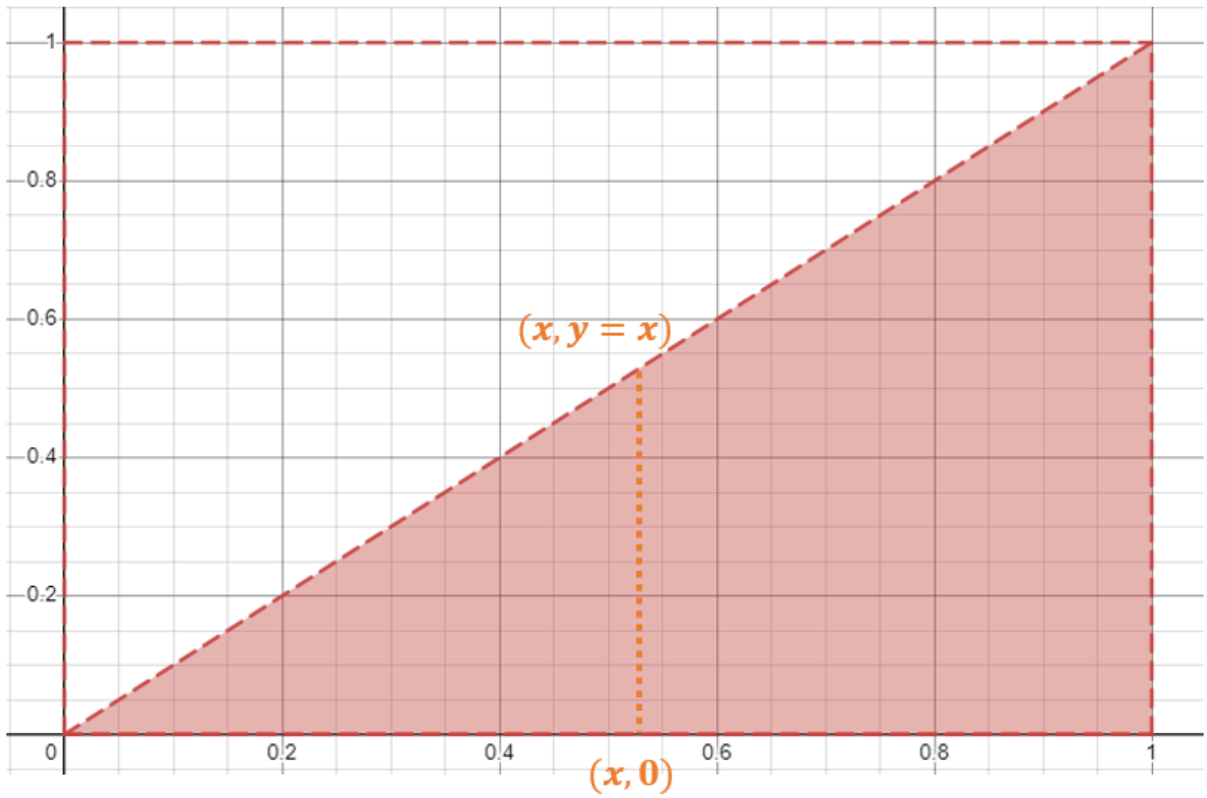
Question 7

M2L9 M2L11

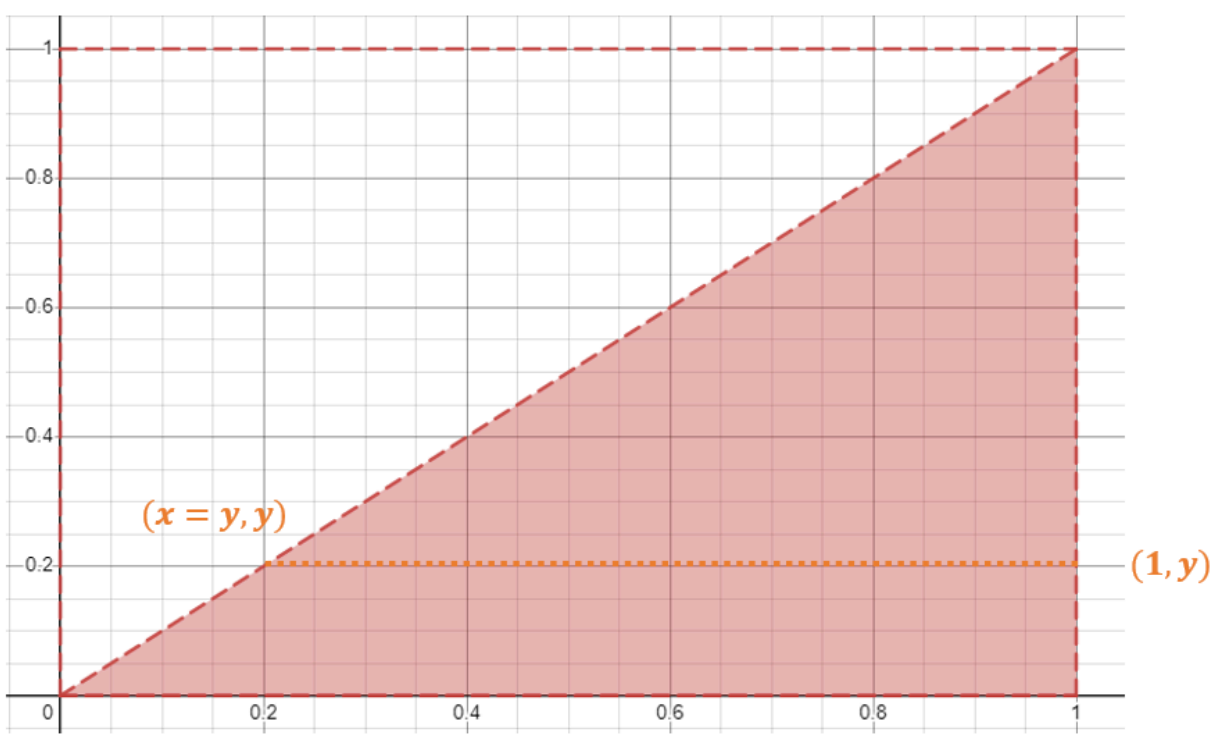
The corresponding region we are integrating over is depicted below; also see [this graph on Desmos](Desmos).



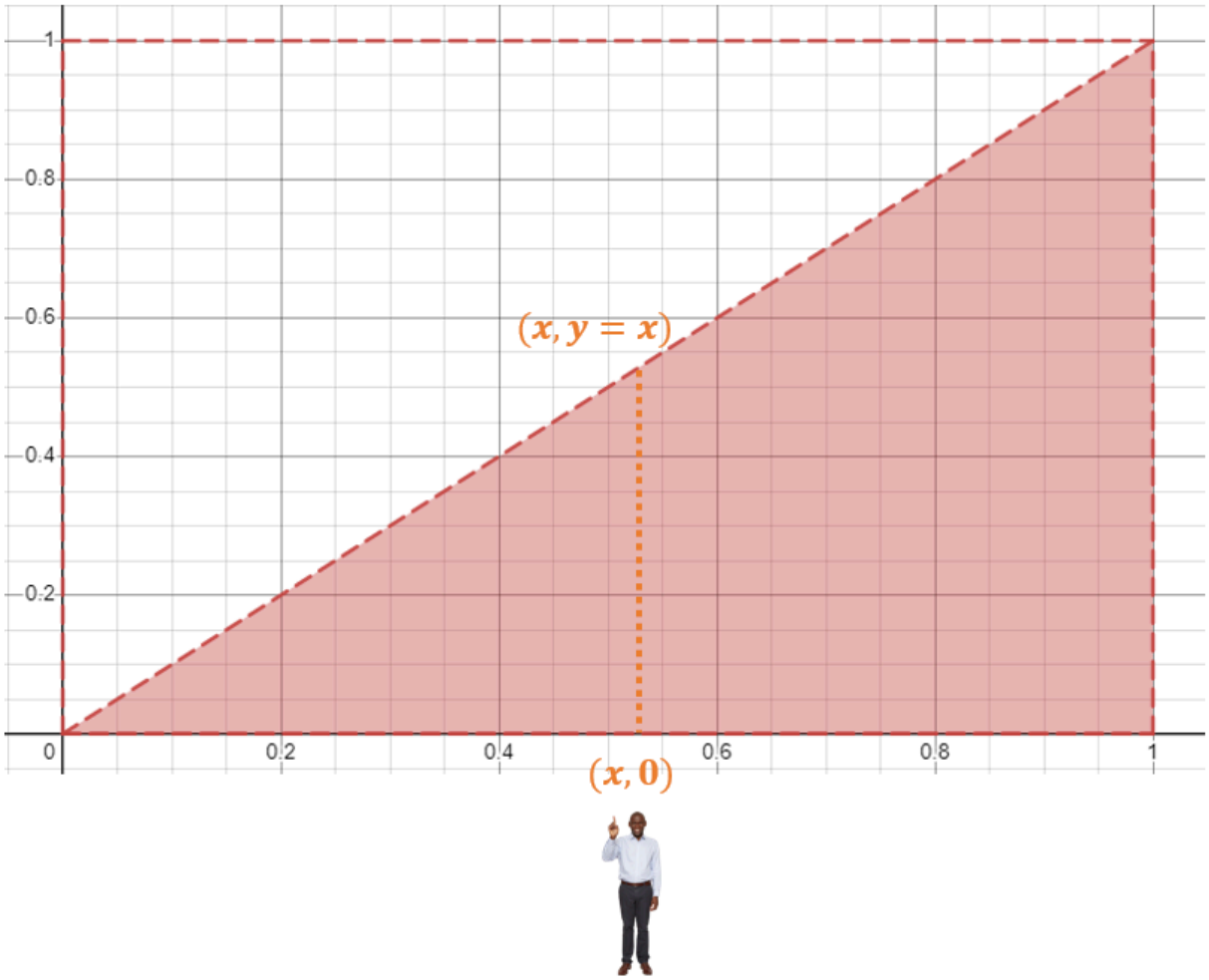
To find the marginal of x, we need to integrate from 0 to $y = x$



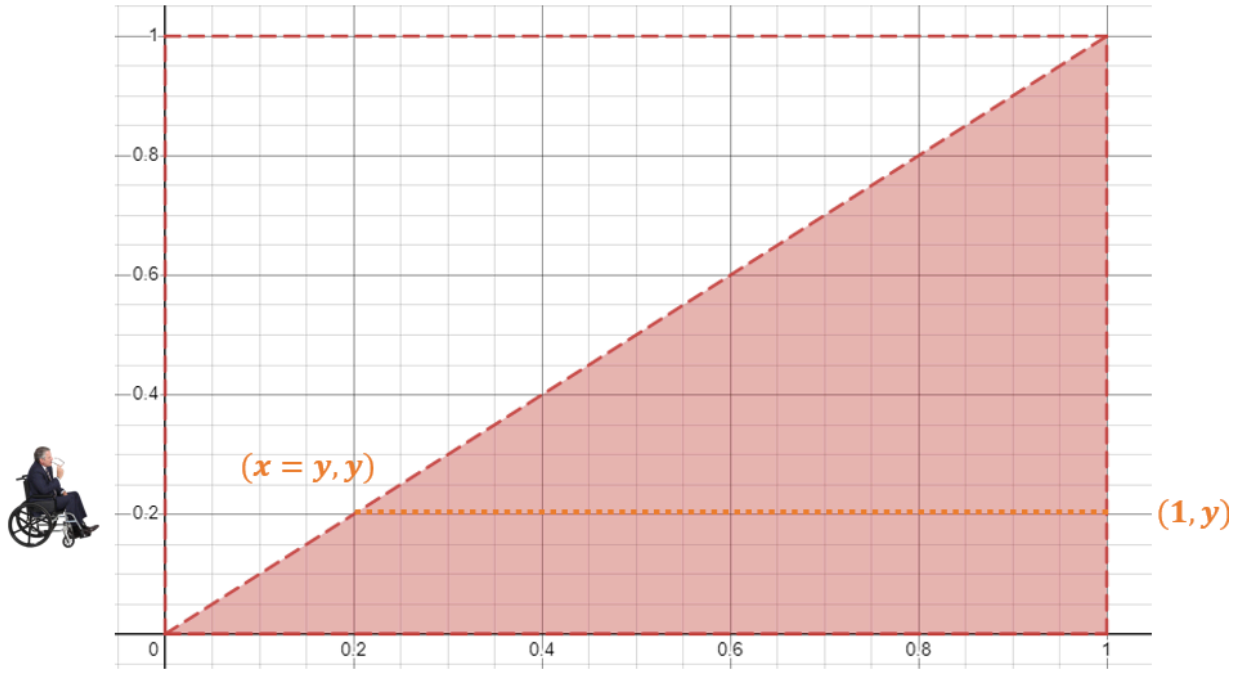
To find the marginal of y, we need to integrate from $x = y$ to 1.



For finding $E[XY]$, the solution shows integrating in the order $dydx$. This corresponds with the graph below. Imagine standing somewhere on the x-axis and looking up. You will integrate from the x-axis up to the line $y = x$.



Integrating in the order $dxdy$ corresponds with the image below. You will integrate from the line $x = y$ to the line $x = 1$.



$$\int_0^1 \int_y^1 8x^2y^2 dxdy = \int_0^1 8y^2 \int_y^1 x^2 dxdy = \int_0^1 8y^2 \left[\frac{x^3}{3} \bigg|_y^1 \right] dy$$

$$= \int_0^1 8y^2 \left(\frac{1}{3} - \frac{y^3}{3} \right) dy = \int_0^1 \frac{8y^2}{3} - \frac{8y^5}{3} dy = \frac{8y^3}{9} - \frac{8y^6}{18} \bigg|_0^1$$

$$= \frac{8}{9} - \frac{8}{18} = \frac{8}{9} - \frac{4}{9} = \frac{4}{9}$$

Question 8

M2L12

Recall from the lecture slides the following:

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, then $Y \equiv \sum_{i=1}^{n} X_i \sim \text{Gamma}(n, \lambda)$. The $\text{Gamma}(n, \lambda)$ is also called the $\text{Erlang}_n(\lambda)$. It has cdf

$$F_Y(y) = 1 - e^{-\lambda y} \sum_{j=0}^{n-1} \frac{(\lambda y)^j}{j!}, \quad y \geq 0.$$

Since X and Y are iid exponential with $\lambda = 1$, the random variable $Z = X + Y$ is an Erlang random variable with $n = 2$ and $\lambda = 1$

The CDF of Z is

$$F_Z(z) = P(Z \leq z) = 1 - e^{-\lambda z} \sum_{j=0}^{n-1} \frac{(\lambda z)^j}{j!}$$

We want to find $P(Z \geq 2)$ for which we will use the complement of the CDF $1 - P(Z \leq 2)$

$$P(Z \geq z) = 1 - P(Z \leq z) = 1 - \left[1 - e^{-\lambda z} \sum_{j=0}^{n-1} \frac{(\lambda z)^j}{j!}\right] = e^{-\lambda z} \sum_{j=0}^{n-1} \frac{(\lambda z)^j}{j!}$$

Note that the use of $\geq$ and $\leq$ is somewhat arbitrary here as it will not have an impact on the results since an Erlang random variable is a continuous random variable.

Using $\lambda = 1$ and $n = 2$:

$$P(Z \geq z) = e^{-z} \sum_{j=0}^{2-1} \frac{z^j}{j!}$$

$$P(Z \geq 2) = e^{-2} \sum_{j=0}^{1} \frac{2^j}{j!} = e^{-2} \left[\frac{2^0}{0!} + \frac{2^1}{1!}\right] = e^{-2}(2+1) = 3e^{-2}$$

Question 9

M2L12

If you have a calculator that computes probabilities for Normal distributions with arbitrary mean and variance/standard deviation, you can do the calculation directly.

Otherwise, the random variable must be standardized and the standard Normal table should be used.

$$P(X \geq 4) = P\left(\frac{X-\mu}{\sigma} \geq \frac{4-1}{\sqrt{9}}\right) = P(Z \geq 1)$$

Review the document [How to use the Normal/t/χ^2 Tables]

We will use the complement to find the probability.

$$P(Z \geq 1) = 1 - P(Z \leq 1)$$

To find $P(Z \leq 1)$:

2

Standard Normal Table

1 3

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	[illegible]	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	[illegible]	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	[illegible]	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

$$P(Z \geq 1) = 1 - P(Z \leq 1) = 1 - 0.8413 = 0.1587$$

You could also get a ballpark figure for this by knowing the 68-95-99.7 Rule. Since the probability is 0.68 is within 1 standard deviation this means there is $1 - 0.68 = 0.32$ that falls either to the left of -1 and to the right of 1. Since the Normal distribution is symmetric, this means half the area falls to the left and half to the right. $0.32/2 = 0.16$ which is close to the result we found above of 0.1587

Question 10

M2L7 M2L12

The sum of two iid Normal random variables is also a Normal random variable.

Recall $E[X+Y] = E[X] + E[Y]$, $E[aX+b] = aE[X] + b$,
$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ (since they are iid), and $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$

$E[X] = E[Y] = 0$ and $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$

We are interested in $P(Y < 2X+1)$ so we can rewrite this as $P(Y - 2X < 1)$

You can think of $Y - 2X$ as a new random variable in which we need to find its mean and variance.

$$E[Y - 2X] = E[Y] - 2E[X] = 0 - 2(0) = 0$$

$$\mathrm{var}(Y - 2X) = \mathrm{Var}(Y) + 2^2\mathrm{Var}(X) = 1 + 4 = 5$$

The random variable $Y - 2X \sim \mathrm{Nor}(0, 5)$

Calculating the probability can be done similarly to Question 9

$$P\left(\frac{(Y-2X) - \mu}{\sigma} < \frac{1-0}{\sqrt{5}}\right)$$

$$\frac{1}{\sqrt{5}} \approx 0.447$$

The probability is somewhere between 0.6700 and 0.6736. You can think of it as being 70% of the way between 0.44 and 0.45.

Standard N al Table

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.73	.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.77	.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

$0.6736 - 0.6700 = 0.0036$; 70% of this is 0.00252

$0.6700 + 0.00252 = 0.67252$

Thus, $P\left(Z \leq \frac{1}{\sqrt{5}}\right) \approx 0.67252$

Question 11

M2L11

Recall that for two random variables, $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$

$\mathrm{Var}(X) = \mathrm{Var}(Y) = 10$ and $\mathrm{Cov}(X,Y) = 6$

$\text{Var}(X+Y) = 10 + 10 + 2(6) = 32$

Question 12

On some of the practice exams, there were questions related to pop culture, which were considered bonus questions. Please do not stress over these, as they are meant to be fun and easy free points. If questions such as these appear on actual exams, you will get credit for them regardless.

Question 13

M2L12

We are looking to find the number of games the Hawks have to play (x) such that their probability of winning 1 game is greater than 0.9. They could do this by winning the first game (W), the second game (LW), the third game (LLW), etc. (W = win, L = loss)

Here is a method without using the CDF.

The PMF is $P(X = x) = (0.4)^{x-1}(0.6)$

We will accumulate the probabilities from the PMF until the probability is ≥ 0.9

$P(X = 1) = (0.4)^0(0.6) = 0.6$

$P(X = 1) + P(X = 2) = 0.6 + (0.4)^1(0.6) = 0.6 + 0.24 = 0.84$

$P(X = 1) + P(X = 2) + P(X = 3) = 0.6 + 0.24 + (0.4)^2(0.6) = 0.936$

A probability ≥ 0.9 is achieved when $x = 3$.

We can also solve this using the CDF and algebra:

$1 - (0.4)^x \geq 0.9$

$0.1 \geq (0.4)^x$

$\ell n(0.1) \geq x\ell n(0.4)$

When dividing by $\ell n(0.4)$, the inequality sign is reversed since $\ell n(0.4)$ is a negative number.

$$\frac{\ell n(0.1)}{\ell n(0.4)} \leq x$$

.$x \geq 2.52$ but since a geometric random variable is discrete, we must round up to 3

Question 14

M2L13

This is a direct result from the Central Limit Theorem.

Question 15

M2L13

The PMF of the random variable is:

$$f(x) = \begin{cases} 0.5 & x = -1 \\ 0.5 & x = 1 \end{cases}$$

The expected value of X is $E[X] = \sum_x x f(x) = (-1)(0.5) + (1)(0.5) = 0$

The variance of X is $E[X^2] - (E[X])^2$

$$E[X^2] = \sum_x x^2 f(x) = (-1)^2(0.5) + (1)^2(0.5) = 1$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = 1 - (0)^2 = 1 - 0 = 1$$

The CLT tells us that the approximate distribution of the sum of 100 of these random variables will be $\mathrm{Nor}(0, 100)$

The 0 comes from adding 0 100 times and the 100 comes from adding 1 100 times.

We can now find the probability:

$$P\left(\frac{\sum_{i=1}^{100} X_i - \mu}{\sigma} > \frac{10-0}{\sqrt{100}}\right) = P(Z > 1)$$

Review Question 9 for the probability calculation.

Question 16

M1L8 M2L6 M3L8

$X_{i+1} = (3X_i + 1) \mod 8$ with $X_0 = 1$

$X_1 = (3X_0 + 1) \mod 8 = 4 \mod 8 = 4$

$U_1 = 4/8 = 0.5$

To find U_{801}, we need to see what the cycle length of this LCG is.

$X_2 = 13 \mod 8 = 5$
$X_3 = 16 \mod 8 = 0$
$X_4 = 1 \mod 8 = 1$

So, the LCG starts repeating after 4 iterations.

This implies that $X_{801} = X_{797} = \cdots = X_5 = X_1$

A more direct way is to take $801 \mod 4 = 1$, so we use the same value as X_1.

Question 17

M1L8 M2L6 M3L8

Recall that $-\frac{1}{\lambda}\ell n(1-U) \sim -\frac{1}{\lambda}\ell n(U) \sim \text{Exp}(\lambda)$

To find the value of λ,

$$-\frac{1}{\lambda} = -3$$

$$1 = 3\lambda$$

$$\lambda = \frac{1}{3}$$

You could also start from

$X = -3\ell n(1-U)$ and solve for U which is equivalent to the random variable $F(X)$ which will give us the CDF of the random variable.

$$-\frac{1}{3}X\ell n(1-U)$$

$$e^{-\frac{1}{3}X} = 1 - U$$

$$U = 1 - e^{-\frac{1}{3}X}$$

This is the CDF of an exponential random variable with $\lambda = 1/3$

Question 18

M1L8 M2L6 M3L8

Using properties of logarithms:

$$-3\ell n(U_1U_2) = -3(\ell n(U_1) + \ell n(U_2)) = -3\ell n(U_1) + -3\ell n(U_2)$$

Recall that $-\frac{1}{\lambda}\ell n(1-U) \sim -\frac{1}{\lambda}\ell n(U) \sim \mathrm{Exp}(\lambda)$

Each of these is an exponential random variable with $\lambda = 1/3$

Adding these results in an Erlang distribution with $n = 2$ and $\lambda = 1/3$

Question 19

M2L6 M2L7

This is a Box-Muller transformation.

$\sqrt{-2\ell n(U_1)}\cos(2\pi U_2) \sim \mathrm{Nor}(0,1)$

$2 + \mathrm{Nor}(0,1)$ is what we are looking for.

Let $X \sim \mathrm{Nor}(0,1)$

$E[2+X] = E[2] + E[X] = 2 + 0 = 2$

$\mathrm{Var}(2+X) = \mathrm{Var}(X) = 1$

The resulting random variable is $\sim \mathrm{Nor}(2,1)$

Question 20

M3L2

$\frac{1}{4}\sum_{i=1}^{4}[1 + cos(\pi U_i)]$

$= \frac{1}{4}[(1 + cos(\pi 0.419)) + (1 + cos(\pi 0.109)) + (1 + cos(\pi 0.732)) + (1 + cos(\pi 0.893))]$

$\approx \frac{1}{4}[1.252 + 1.942 + 0.334 + 0.056] = \frac{1}{4}(3.584) = 0.896$

If you're using a calculator, make sure your calculator is in radians mode.

Question 21

M3L1

f(x+h)	x	h	f(x)[1+h(x-2)]	
f(0)	0	-	-	1
f(0+0.01)	0	0.01	f(0)[1+(0.01)(0-2)]=1[1 -0.02]	0.98
f(0.01+0.01)	0.01	0.01	f(0.01)[1 + (0.01)(0.01-2)]=0.98[1-0.0199]	0.960498

Question 22

M3L5

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
1	8				3*6=18	
2	2				3*2=6	
3	5				3*3=9	
4	2				3*1=3	

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
1	8	8	8	0	18	26
2	2				6	
3	5				9	
4	2				3	

The first customer leaves at time 26.

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
1	8	8	8	0	18	26
2	2	10			6	
3	5	15			9	
4	2	17	26	9	3	29

In the first 20 minutes,

Customer 1 has been there since time 8; 20 - 8 = 12
Customer 2 has been there since time 10; 20 - 10 = 10
Customer 3 has been there since time 15; 20 - 15 = 5
Customer 4 has been there since time 17; 20 - 17 = 3

12 + 10 + 5 + 3 = 30

Another way of thinking about this is to look at the number of customers in the system in each interval.

[8, 10]: Customer 1 (1)
[10, 15]: Customers 1 and 2 (2)
[15, 17]: Customers 1, 2, and 3 (3)
[17, 20]: Customers 1, 2, 3, and 4 (4)

1(10 - 8) + 2(15 - 10) + 3(17 - 15) + 4(20 - 17)=2+10+6+12=30

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
1	8	8	8	0	18	26
2	2	10			6	
3	5	15	29	14	9	38
4	2	17	26	9	3	29

i	I_i	A_i	T_i	W_i^Q	S_i	D_i
1	8	8	8	0	18	26
2	2	10	38	28	6	44
3	5	15	29	14	9	38
4	2	17	26	9	3	29

Joey will make $2(6 + 3 + 2 + 1) = 24$

Joey will lose $0.5(0 + 28 + 14 + 9) = 25.5$

$24 - 25.5 = -1.5$